

EPREUVE DU 20 JANVIER 2020

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space. Let $T > 0$. Let (W_t) be a standard $\mathbb{R}$ -valued $(\mathcal{F}_t)$ -Brownian motion, where $(\mathcal{F}_t)$ is a filtration fulfilling the usual conditions.

1. Let $\phi \in C^1(\mathbb{R})$ and that X is a continuous semimartingale with respect to $(\mathcal{F}_t)$ with martingale component M . Explain why $\phi(X_t)$ is an $(\mathcal{F}_t)$ -Dirichlet process.
2. Let us consider a linear operator $\mathcal{L}$ on algebra $\mathcal{D}$ (linear subspace stable by the multiplication) of continuous functions $[0, T] \times \mathbb{R}$. In particular

$$\mathcal{L} : \mathcal{D} \subset C([0, T] \times \mathbb{R}) \rightarrow C([0, T] \times \mathbb{R}).$$

Let $X = (X_t)_{t \in [0, T]}$ be an $\mathbb{R}$ -valued continuous $(\mathcal{F}_t)$ -adapted process. Suppose that X solves the **pseudo-martingale problem** with respect to $(\mathcal{D}, \mathcal{L})$ in the sense that

$$M^f := f(t, X_t) - f(0, X_0) - \int_0^t (\mathcal{L}f)(r, X_r) dr, \quad (1)$$

is an $(\mathcal{F}_t)$ -local martingale, for every $f \in \mathcal{D}$.

- (a) Let $f \in \mathcal{D}$. Explain why $(f(t, X_t))$ is a (continuous) semimartingale for every $f \in \mathcal{D}$.
- (b) We define the following bilinear map $\mathcal{B} : \mathcal{D} \times \mathcal{D} \rightarrow C([0, T] \times \mathbb{R})$:

$$\mathcal{B}(f, g) := \mathcal{L}(fg) - f\mathcal{L}(g) - g\mathcal{L}(f).$$

Using (1) and using integration by parts show that

$$[f(\cdot, X), g(\cdot, X)]_t = M_t + \int_0^t \mathcal{B}(f, g)(r, X_r) dr, \quad (2)$$

where M is a local martingale.

- (c) Let $f, g \in \mathcal{D}$. Explain why $[f(\cdot, X), g(\cdot, X)]$ is a bounded variation process. Deduce from (2) the expression of $[f(\cdot, X), g(\cdot, X)]$.
- (d) Let $a, b : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ are continuous functions. What is the meaning of existence in law of $E(a, b)$?
- (e) Suppose that X is solution to

$$X_t = X_0 + \int_0^t a(s, X_s) dW_s + \int_0^t b(s, X_s) ds. \quad (3)$$

We set $\mathcal{D} := C^{1,2}([0, T] \times \mathbb{R})$. Determine a map $\mathcal{L}$ such that that X verifies the pseudo-martingale problem with respect to $(\mathcal{D}, \mathcal{L})$.

- (f) Determine the map $\mathcal{B}$ in previous case and consequently $[f(\cdot, X), g(\cdot, X)]$ for $f, g \in \mathcal{D}$.
- (g) We set $\mathcal{D} = C^2(\mathbb{R})$. Let X be a process solving the pseudo-martingale problem with respect to $\mathcal{D} = C^2(\mathbb{R})$, $\mathcal{L}f = \frac{f''}{2}$ for every $f \in C^2$.
- Show (X_t) and (X_t^2) are semimartingales and provide their decompositions.
 - Deduce that $Y_t = X_t - X_0, t \in [0, T]$ is a Brownian motion.
3. **Generalized FBSDEs.** Suppose that $id \in \mathcal{D}$ where $id(x) = x$. Clearly X is an $(\mathcal{F}_t)$ -semimartingale and denote by M^X the $(\mathcal{F}_t)$ -local martingale component. Let $g : [0, T] \times \mathbb{R}^3 \rightarrow \mathbb{R}, \phi : \mathbb{R} \rightarrow \mathbb{R}$ be continuous. We say that a triple (Y, Z, O) of progressively measurable continuous processes is a solution of $GFBSDE(g, \phi)$ if the following holds.
- O is an $(\mathcal{F}_t)$ -local martingale vanishing at zero.
 - $$Y_t = \phi(X_T) - \int_t^T Z_s dM_s^X + \int_t^T g(s, X_s, Y_s, Z_s) ds - (O_T - O_t), t \in [0, T].$$
 - $[O, M^X] = 0$.
- Let (Y, Z, O) be a solution of $GFBSDE(g, \phi)$. Show that $[Y, M^X]_t = \int_0^t Z_s d[X]_s, t \in [0, T]$.
 - Let us suppose that X solves the pseudo-martingale problem with respect to some $(\mathcal{D}, \mathcal{L})$. Let $u : [0, T] \times \mathbb{R} \rightarrow \mathbb{R} \in \mathcal{D}$ such that $\mathcal{L}u(t, x) = g(t, x, u(t, x), v(t, x))$ with $\mathcal{B}(id, id)v = \mathcal{B}(u, id)$ and $id(x) = x$. Observe that $[M^X]_t = \int_0^t \mathcal{B}(id, id)(X_s) ds, t \geq 0$. Show the existence of a process O such that (Y, Z, O) is a solution of $FBSDE(g, \phi)$ with $Y_t = u(t, X_t), Z_t = v(t, X_t)$.
 - Suppose that $X = W$. Suppose that (Y, Z) is a solution of the classical forward-backward $SDE(g, \phi, 1, 0)$. Provide a solution to the $GFBSDE(g, \phi)$.
4. Let $b : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}_+$ be a Lipschitz non-negative function, $x_0, a \geq 0$. We consider the SDE
- $$\begin{cases} dX_t &= a\sqrt{|X_t|}dW_t + b(t, X_t)dt \\ X_0 &= x_0. \end{cases} \quad (4)$$
- Recall the concept of pathwise uniqueness of an SDE.
 - Explain why previous SDE (4) admits pathwise uniqueness.
 - Explain why a solution X of (4) is necessarily non-negative.
 - Suppose that X is a strictly positive solution of (4). Determine the SDE solved by $Y_t := \sqrt{X_t}, t \geq 0$.
 - Specify a and b such that a solution X of (4) is a square Bessel process.
5. Let X be a Bessel process of the type $BES(x_0, \delta)$ with $x_0 = 0$ and $\delta = 1$. Determine $E(X_t^3)$ for every $t \geq 0$.

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Advanced Stochastic Calculus (Second part) (FQ308-OMI302). END